Full Derivation: The Functional Reality Transfer Pipeline (FRPT)

1. Foundational Action Principle

We define the universal action I_U as the variational minimum of network entropy and structural tension:

$$\delta \int [\, Tr(\rho \ln \rho) + \gamma \int d^4x \sqrt{|g|} \, (1/(16\pi G) R + 1/(4g^2) Tr(F_{\mu\nu}F^{\mu\nu})) + \beta \sum_{i,j} W_{ij} D_{KL}(q_i \,||\, p_j) \,]\, dt = 0$$

2. Emergence of Spacetime Geometry

The metric $g_{\mu\nu}$ is derived via the heat kernel expansion of the discrete Graph Laplacian L_G:

$$Tr(e^{-t\, L_G}) \sim 1/(4\pi t)^{d/2} \sum t^m \int a_m(x) \sqrt{|g|}\, d^d x$$

Minimizing the discrete Laplacian trace $Tr(L_G)$ is isomorphic to minimizing the Einstein-Hilbert action in the continuum limit.

3. Gauge Fields as Network Holonomies

The Yang-Mills field strength arises from the plaquette action of the graph's link variables U_{ij}:

$$C_{gauge} = 1/g^2 \sum_{\square} Re\, Tr(I - U_{\square}) \Rightarrow 1/(4g^2) \int Tr(F_{\mu\nu}F^{\mu\nu}) \sqrt{|g|}\, d^4x$$

4. Topological Mass and Coupling

The proton-to-electron mass ratio μ is derived as the ratio of the structural volume required for topological stabilization:

$$\mu = m_p / m_e \approx (N_p / N_e)^{3/2}$$

The fine-structure constant $\boldsymbol{\alpha}$ is similarly derived as a topological invariant of the graph's trivalent connectivity density.